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Vitual Reality Prism Effect

Group 1 — 8. Semester project

Authors

Markus Birch Flensburg

Mathias Bisgaard

Rikke Bragh Jensen

Tze Huo Gucci Ho

Supervisors

Bastian Ilsø Hougaard

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Project group:

Markus Birch Flensburg

mflens22@student.aau.dk

Mathias Bisgaard

mbisga22@student.aau.dk

Rikke Bragh Jensen

rbje22@student.aau.dk

Tze Huo Gucci Ho

tho22@student.aau.dk

Abstract:



Supervisors: Bastian Ilsø Hougaard
Department: Department of Architecture, Design and Media Technology
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Introduction

Background Research

2.1 Spatial neglect in stroke victims

Unilateral spatial neglect (USN) is a common consequence following a stroke and is characterised by a reduced ability to detect, respond to, or orient toward stimuli presented on the contralesional side of the brain lesion[9]. The disorder reflects a deficit in spatial attention and spatial representation rather than a primary sensory or motor impairment. Consequently, patients may neglect and fail to acknowledge objects, people, or even their own limbs on the affected side, reflecting deficits across both egocentric and allocentric spatial representations, significantly interfering with activities of daily living (ADL) and impeding recovery. Because USN affects both perceptual and motor aspects of spatial behaviour, rehabilitation methods must target the pathological attentional bias toward the ipsilesional side rather than isolated sensory or motor symptoms. To evaluate these deficits, researchers typically rely on behavioural measures, which are observable performance on spatial tasks that probe different components of spatial processing. Common examples include line bisection, which assesses perceived midline[7, 4, 12]; landmark task, which measures spatial judgement of pre-bisected lines; and open-loop pointing, which evaluates perceived straight-ahead without visual feedback[3, 11, 1, 5, 2]. These methods will be elaborated further in Section 3.1.3.

2.2 Prism therapy

A well-established method for rehabilitating unilateral spatial neglect is prism adaptation therapy, in which patients typically perform repeated goal-directed visuomotor tasks while wearing laterally displacing wedge prism glasses[14, 4, 15]. The prisms are configured to shift the visual field toward the ipsilesional side of the brain lesion, thereby creating a systematic mismatch between visual input and proprioceptive-motor representations. During task performance, this visuomotor mismatch initially produces expected pointing errors, but continued exposure over repeated trials eventually elicits automatic, error-driven sensorimotor recalibration through repeated movement corrections[13]. Upon removal of the prisms, a compensatory after-effect emerges in the direction opposite to the optical displacement. Because the initial displacement is ipsilesional, the after-effect is expressed toward the contralesional side of space, counteracting the pathological spatial bias characteristic of neglect[14, 4, 15]. Importantly, the therapeutic efficacy of prism adaptation depends not only on this low-level sensorimotor recalibration as reflected in the after-effect, but also on its transfer to higher-order spatial processing and activities of daily life. Standard prism therapy procedures generally involve the following three phases:

1. A pre-exposure phase where baseline visuomotor performance is assessed on a rehabilitation task prior to any prism effect being introduced. This phase establishes the patient's initial spatial and motor accuracy, providing a reference for evaluating subsequent prism adaptation.
2. An exposure phase where patients perform repeated goal-directed actions under prism exposure, during which the prism effect introduces a systematic visuomotor mismatch. Initial pointing errors occur due to this mismatch, where repeated movements eventually lead to

gradual error reduction as the sensorimotor system undergoes recalibration to compensate for the visuomotor mismatch.

3. A post-exposure phase in which prism exposure ends and the prisms are removed. The resulting after-effect of sensorimotor recalibration is then evaluated, and visuomotor performance is reassessed on the same rehabilitation task used in the pre-exposure phase. This phase also examines the transfer of the after-effect and sensorimotor recalibration to higher-order spatial processes.

Prism adaptation (PA) refers to the combined process encompassing error reduction and sensorimotor recalibration during prism exposure, as well as the subsequent after-effect and its transfer once the prisms are removed.

2.3 Prism therapy in virtual reality

Rapid advances in immersive technologies have positioned virtual reality (VR) as a promising clinical tool for neurorehabilitation by enabling an engaging, controlled, and highly configurable therapeutic environments.[6] In recent years, VR has been explored as a method not only for facilitating prism therapy but also for supporting a wider range of interventions for spatial neglect rehabilitation. VR-based implementations of prism therapy generally follow the same procedure as conventional prism therapy, but where the prism effect is simulated digitally rather than produced through physical wedge prism glasses.[3]

[10]

Compared to conventional prism therapy, VR offers several potential strengths. First, immersive head-mounted displays can reduce external visual and auditory distractions while providing a controlled visual environment. Second, VR allows the perturbation to be implemented and adjusted digitally, enabling precise control over the magnitude and direction of the induced prism effect. Third, the virtual environment allows task parameters such as target position, timing, and feedback to be adjusted in detail while simultaneously enabling streamlined data collection. These capabilities allow the full prism adaptation process to be monitored on a trial-by-trial basis rather than relying solely on pre-post measurements. For example, immediate visual feedback of the pointing endpoint provides a direct and quantifiable measure of motor accuracy, allowing the temporal evolution of error reduction and sensorimotor recalibration to be analysed throughout prism exposure. The controlled nature of VR environments may also reduce awareness of the spatial manipulation, which has been suggested to increase the likelihood of successful adaptation in some studies.[10][3][6]

Despite these advantages, VR-based prism therapy also presents potential limitations. Firstly, VR technology can inherently induce cybersickness symptoms in some users, which may cause discomfort and potentially reduce therapeutic efficacy[8]. In addition, the sensorimotor experience in VR differs from real-world interactions due to factors such as tracking accuracy, display latency, and differences in depth perception. As a result, it remains important to establish whether VR-induced perturbations produce prism adaptation comparable to those observed with physical prism glasses.[10][3][6]

Several studies have examined this question by comparing prism adaptation tasks performed with traditional prism glasses and in immersive VR environments. Studies have indicated that VR can induce adaptation in a comparable or even larger proportion of participants while producing after-

effects of similar magnitude to conventional prism therapy[10][6]. At the neural level, immersive VR-based prism therapy has also been associated with activation of cortical areas involved in sensorimotor processing and attentional control. Together, these findings suggest that VR implementations can reproduce key aspects of conventional prism adaptation, supporting their use as a controllable platform for inducing and measuring prism-adaptation-related spatial recalibration.[3]

2.4 State of the Art for Prism Therapy in Virtual Reality

The current state of the art in virtual reality implementations of prism therapy present a wide variety of approaches. Across the literature, implementations differ in (1) how the perturbation is generated (e.g., optical prisms, virtual hand offset or rotation, environment/target manipulations), (2) how feedback is presented (e.g., terminal feedback only or continuous feedback; explicit error visualisation), and (3) which tasks were employed for performance assessment (e.g., open-loop pointing, line or landmark bisection, head-orientation metrics). To provide a detailed overview of current approaches, a structured literature review was conducted. The following sections summarise its key findings, with the full review available in Appendix X.

2.4.1 Comparing Conventional Prism Therapy to a Virtual Reality Approach with Open-Loop Pointing

Anan et al. directly compared conventional prism glasses with a virtual reality-based prism adaptation (VR-PA) system employing an open-loop pointing test in both conditions, making their study a useful foundation for understanding what VR-PA means operationally. They fixed the deviation to 20 diopters (approximately 11.3°) rightward in both the prism-glasses and VR conditions, and used a repeated reaching protocol of 96 reaches to three targets arranged in an arc[1]. For the conventional prism glasses, reaching was performed from below a table towards tabletop targets, paced by a metronome every 3 seconds[1]. In the VR condition (Oculus Rift + Touch controllers; Unity), the same structure was recreated with three spherical targets in an arc; every 3 seconds one target changed colour as a cue, and participants pressed a controller button where they believed the target was[1]. The VR system explicitly visualised endpoint error (e.g., different colours depending on whether the reach overlapped the target), and the authors emphasise the advantage in VR which enables fine-grained adjustment of the deviation angle in 1 diopter increments and logging of performance data[1]. Their operational definition of successful adaptation was a leftward shift on the open-loop pointing test after exposure.[1]

2.4.2 Virtual prism adaptation therapy via virtual-hand displacement and neurophysiological data collection

Cho et al. developed an immersive virtual prism adaptation therapy (VPAT) system that induces the prism effect by shifting the virtual hand relative to the real hand during repeated pointing, explicitly aiming to replicate conventional prism therapy without physical prisms[3]. In their feasibility study with 14 healthy adults, the protocol consisted of four sequential phases: pre-VPAT, VPAT- 10° , VPAT- 20° , and post-VPAT[3]. The key prism adaptation signature is described in similar terms as conventional prism therapy: early exposure shows rightward pointing errors under the imposed deviation, errors decrease with repeated pointing (error reduction), and a leftward after-effect appears in the post-exposure phase when the deviation is removed[3]. Methodologically, the study's distinct contribution is that it couples the behavioural protocol to functional near-infrared spectroscopy (fNIRS), where they analyse changes in oxyhaemoglobin concentration during pointing

across phases and report that the most significantly activated channels during pointing contrasts were located in the right hemisphere, with candidate regions including dorsolateral prefrontal cortex and frontal eye field[3]. Their interpretation is that VPAT can induce behavioural adaptation and may modulate attention-related networks while providing an instrumented way to quantify errors in VR.[3]

2.4.3 Visuomotor Adaptation as Spatial Recalibration with Virtual-Hand Rotation Deviation

Gerken et al. explicitly frame prism adaptation in VR within the broader concept of visuomotor adaptation: a motor-learning process driven by sensory prediction errors under a systematic mismatch between intended movement and observed feedback[6]. In their VR task, the perturbation is implemented as a 40° rotation of visual feedback during repeated reaching movements[6]. Rather than varying the deviation magnitude across tasks, the manipulation targets the format of feedback, which is an embodied virtual hand versus an abstract cursor, to test whether embodiment and the associated sense of agency changes the adaptation process[6]. Behaviourally, they report that hand-based feedback speeds up early error reduction, whereas after-effects and transfer measures are not altered by feedback type, which they interpret as evidence that the underlying recalibration mechanism is not determined by visual representation alone[6]. This study is representative of a class of VR approaches that induce spatial recalibration via visuomotor rotation (a controlled mismatch in feedback mapping) rather than a shift in the visual field.

2.4.4 Virtual Hand Displacement for Reproducing PA without Prisms

Kim et al. propose that the key component to PA is the repeated coupling of movement to systematically displaced visual feedback, not the optical prism device itself[10]. Their VR approach induces the prism effect by applying an angular deviation between the tracked physical hand and the displayed virtual hand during repeated pointing, such that exposure produces systematic reaching errors that reduce over time and a measurable after-effect once the deviation is removed[10]. In this interpretation, VR is primarily a platform for parametrising the deviation (e.g., selecting the deviation magnitude and logging trial-by-trial error) while keeping the broader visual environment stable[10]. This aligns with later VR-PA comparisons that treat prism effect as an imposed mapping error between action and visual feedback rather than a global shift of the visual scene.[1, 3]

2.4.5 Multisensory Re-orientation of Attention in Immersive Rehabilitation

Heyse et al. demonstrate a qualitatively different direction. Their aim was expand conventional prism therapy by addressing multiple modalities instead of just optical shifts. Rather than inducing PA through a visuomotor discrepancy and measuring after-effects, they design a multisensory VR tool intended to re-orient attention and exploration in patients with unilateral spatial neglect[7]. Their system uses a xylophone-like instrument where key placement can be changed per task. For assessment, keys are arranged symmetrically and for training, keys are placed off-centre towards the neglected side as a bottom-up manipulation intended to draw attention into contralesional space[7]. Over a two-week clinical period, sessions occurred three times per week (Monday/Wednesday/Friday), lasting approximately 30 minutes and consisting of an Assessment followed by Scales, Memory, and Free-to-Play tasks[7]. Behaviour is quantified through task performance measures (e.g., timing/accuracy on keys) and head-direction metrics, with the authors discussing how stimuli presentation and task difficulty adjustments influence exploration towards

the neglected side[7]. This work shows that immersive VR has also been used to target neglect through multisensory cueing and task-driven exploration.[7]

2.4.6 Literature Review of Virtual Reality Treatment of Unilateral Spatial Neglect Using Eye-Tracking Technology

A related work by Kaiser et al. provides a comprehensive literature review of how virtual reality and eye-tracking technologies have been applied in the treatment of unilateral spatial neglect. Their analysis highlights that, although VR and gaze-based interaction offer promising avenues for creating controlled, adaptive rehabilitation environments, current research remains heavily focused on severe or clinically obvious cases of USN. As a result, milder and more heterogeneous forms of neglect, such as subtle attentional biases or deficits that manifest only under specific task demands, are frequently overlooked. Kaiser et al. emphasize that this gap limits the ecological validity and clinical reach of existing VR-based interventions, as these milder USN subtypes are both common and clinically relevant. They therefore call for future work to develop more sensitive assessment methods and design VR tasks capable of addressing the full spectrum of USN presentations. Their findings underline a need for flexible and configurable VR rehabilitation frameworks that can accommodate individual differences in neglect profiles and support targeted therapeutic strategies.[9]

The state-of-the-art review of virtual reality implementations of prism therapy revealed substantial inconsistencies in how researchers describe, simulate, and induce the prism effect and prism adaptation. These inconsistencies extend to terminology, implementation strategies, and task selection. As a consequence, each research group has developed its own interpretation of what the prism effect is, how it should be simulated, and what prism adaptation entails, and has typically implemented a virtual environment specifically tailored to those interpretations. While some methodological elements overlap across studies, these VR environments are generally designed in isolation, vary in structure and parameters, and are seldom built to accommodate different subtypes of unilateral spatial neglect (USN). In practice, this results in research groups reinventing VR solutions for their specific studies, with limited reusability.

This absence of a shared language and framework risks limiting comparability across studies, complicates reproducibility, and restricts the broader applicability of VR-based prism therapy in both research and clinical practice.

Based on these observations, two key research gaps are identified:

1. Lack of a shared terminology: Core concepts such as prism effect and prism adaptation are interpreted and defined differently across studies, complicating interpretability and comparability of findings.
2. Absence of a unified framework: Current VR implementations are largely project-specific and mutually incompatible, offering no standardized or configurable platform for simulating different prism effects, supporting diverse rehabilitation tasks, or accommodating research on USN subtypes. As a result, existing systems vary considerably in structure and functionality, limiting their reusability and comparability across studies.

Collectively, these gaps highlight the need for a shared framework to support future work in VR-based prism therapy. A unified and configurable platform, grounded in consistent terminology and

methodology, would streamline experimentation for rehabilitating USN, allow controlled simulation of different prism effects, and support a broader range of rehabilitation tasks. Such a platform could also enable clinicians and researchers to tailor interventions to specific USN subtypes in a systematic and comparable way.

2.5 Design rationale

The reviewed literature shows that prism-adaptation-like effects can be induced in immersive environments by introducing a systematic mismatch between executed movement and visual feedback. Studies using virtual hand displacement, visually shifted environments, and visuomotor rotation report comparable behavioural signatures in the form of trial-by-trial error reduction during exposure and a compensatory after-effect after removal of the perturbation [10, 1, 3, 6].

2.5.1 Independent variables: induction of prism adaptation

All icons follow a consistent visual convention. Blue elements represent the virtual environment, while grey elements represent the physical (real-world) components. Dotted lines denote raycasts, and red circles indicate target locations. The raycasts are not visible to participants during the experiment but are included in the icons to illustrate the direction and angular deviation of the interaction. The icons are simplified representations designed to emphasize the type of visuomotor manipulation applied.

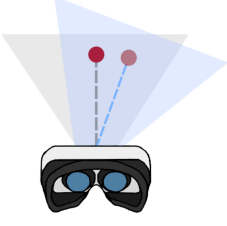
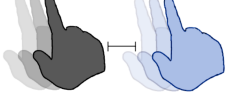
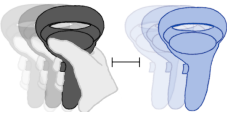
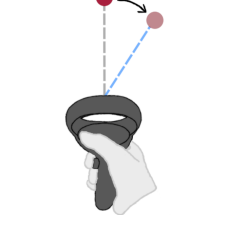
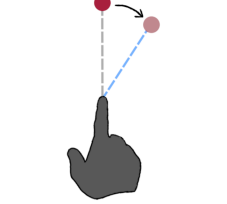
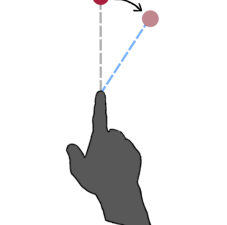
Icon	Name	Description
	Visually shifted environment	A lateral displacement of the visual scene corresponding to an optical deviation similar to that of traditional wedge prism glasses. This condition reproduces the classical prism effect in an immersive environment while allowing precise control of the deviation magnitude[1].
	Virtual hand displacement	A constant lateral offset between the tracked real hand and the rendered virtual hand. This implementation follows VR-based prism-adaptation paradigms in which a positional mismatch produces gradual error reduction and a post-exposure after-effect [10, 1].
	with Arm	
	with Controller	
	Visuomotor rotation	An angular rotation of the visual feedback relative to the executed movement direction. This manipulation is used to provide a reference condition for comparing different reference frames of adaptation[6].
	with Hand	
	with Arm	

Table 2.1: Prism effect icons

2.5.2 Prism adaptation tasks

The following icons illustrate the behavioural tasks used to assess prism adaptation and its after-effects. Each icon highlights the key interaction and response required from the participant.

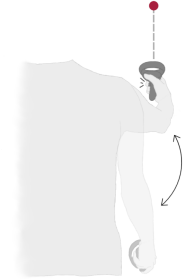
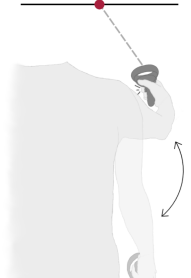
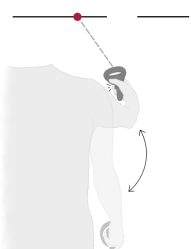
Icon	Name	Description
	Open Loop Pointing	Open-loop pointing is used to measure the after-effect. Participants point towards their perceived straight-ahead without visual feedback of the moving hand. The angular deviation from the true midline indicates the magnitude and direction of the sensorimotor realignment.
	Line Bisection	In the line bisection task, participants are presented with a horizontal line and are asked to indicate its perceived midpoint. The deviation from the true centre provides a measure of spatial bias in the internal representation of space. Changes between pre- and post-exposure performance are interpreted as transfer of the adaptation beyond the trained movement.
	Landmark Bisection	In the landmark task, participants judge whether a pre-bisected horizontal line is transected to the left or right of its true midpoint. Because the response does not require a pointing movement, this task provides a perceptual measure of spatial bias independent of motor execution.

Table 2.2: Task Icons

2.6 Problem statement

Prism therapy is a well-established method for rehabilitating unilateral spatial neglect (USN). Conventional prism therapy setups rely on physical wedge prism glasses and manually administered tasks, which limits experimental flexibility, standardization, scalability, and addressing of different subtypes of USN. Recent research has shown that virtual reality (VR) can effectively be used to create a controllable environment for prism therapy to simulate prism effects and induce prism adaptation for USN rehabilitation. However, existing VR implementations are typically developed for single specific experiments and lack the modularity necessary for enabling broader research or clinical use. To address this gap, we propose the development of a virtual reality sandbox platform that enables researchers and clinicians to design, configure, and employ a wide range of prism effects and rehabilitation tasks, as well as enabling adaptation of interventions to specific USN subtypes. This platform will aim to support experimentation, facilitate adaptable interventions for USN subtypes, and provide a unified environment for prism therapy in VR. In addition, this study

will systematically investigate how different combinations of prism effects and rehabilitation tasks influence the induction of prism adaptation in healthy participants. By evaluating which configurations most effectively elicit prism adaptation, this study aims to gain insights that can guide future therapeutic design. For this purpose, this project will be guided by the following problem statement:

*"How do different VR-based methods for inducing spatial recalibration influence the magnitude and characteristics of behavioural adaptation in healthy participants? OR
How do different implementations of the prism effect in VR affect the magnitude of prism adaptation (including error reduction and after-effect)?"*

To support this problem statement and further guide the design and implementation of this sandbox system, a set of research questions were formulated:

1. How does the prism adaptation implemented in VR affect healthy people in goal-directed pointing tasks?
2. To what extent does the adaptation transfer to spatial tasks that do not require the trained movement?

Methodology

The purpose of the experimental design is to compare different VR-based methods for inducing prism adaptation and to evaluate the resulting after-effects using consistent and measurable outcome variables. The methodology therefore separates the induction of visuomotor mismatch from the procedures used to measure adaptation.

3.1 Design rationale

The reviewed literature shows that prism-adaptation-like effects can be induced in immersive environments by introducing a systematic mismatch between executed movement and visual feedback. Studies using virtual hand displacement, visually shifted environments, and visuomotor rotation report comparable behavioural signatures in the form of trial-by-trial error reduction during exposure and a compensatory after-effect after removal of the perturbation [10, 1, 3, 6].

To ensure methodological consistency across conditions, only induction methods that are expected to produce measurable error reduction and after-effects are included. Approaches that primarily train spatial exploration without generating a sensorimotor realignment are therefore not part of the core comparison.

3.1.1 Independent variables: induction of prism adaptation

Spatial recalibration is induced by manipulating the relation between the physical movement of the participant and its visual representation in VR. The sandbox implements the following visuomotor mismatch conditions:

- **Visually shifted environment:** a lateral displacement of the visual scene corresponding to an optical deviation similar to that of traditional wedge prism glasses. This condition reproduces the classical prism effect in an immersive environment while allowing precise control of the deviation magnitude[1].
- **Virtual hand displacement:** a constant lateral offset between the tracked real hand and the rendered virtual hand. This implementation follows VR-based prism-adaptation paradigms in which a positional mismatch produces gradual error reduction and a post-exposure after-effect [10, 1].
- **Visuomotor rotation:** an angular rotation of the visual feedback relative to the executed movement direction. This manipulation is used to provide a reference condition for comparing different reference frames of adaptation[6].

In addition, the magnitude of the imposed deviation can be introduced progressively across trials. Gradual increases in prism strength have been used in dose-response studies to investigate the temporal development of adaptation and are therefore implemented as a controllable parameter[5].

3.1.2 Procedure: exposure and feedback

Participants perform repeated goal-directed pointing movements towards visual targets. During the exposure phase, one of the prism effect conditions is active.

Two feedback modes are considered:

- **Concurrent feedback**, in which the movement is visible throughout its execution.
- **Terminal feedback**, in which only the endpoint of the movement is shown.

Terminal feedback prevents online correction and is commonly used in prism therapy to promote error-driven realignment, whereas concurrent feedback allows continuous visual guidance of the movement.

3.1.3 Behavioural measures

Behavioural measures are assessed using tasks that capture changes in perceived straight-ahead, spatial bias, and trial-by-trial pointing performance. These measures are selected because they are widely used in prism-adaptation studies and allow direct comparison with previous work.

Pointing error during exposure

During the exposure phase, the angular deviation between the target position and the movement endpoint is recorded for each trial. The reduction of this error across trials is used as a measure of the rate of adaptation.

Open-loop pointing

Open-loop pointing is used to measure the after-effect. Participants point towards their perceived straight-ahead without visual feedback of the moving hand. The angular deviation from the true midline indicates the magnitude and direction of the sensorimotor realignment.

Line bisection task

In the line bisection task, participants are presented with a horizontal line and are asked to indicate its perceived midpoint. The deviation from the true centre provides a measure of spatial bias in the internal representation of space. Changes between pre- and post-exposure performance are interpreted as transfer of the adaptation beyond the trained movement.

Landmark task

In the landmark task, participants judge whether a pre-bisected horizontal line is transected to the left or right of its true midpoint. Because the response does not require a pointing movement, this task provides a perceptual measure of spatial bias independent of motor execution.

Error reduction during exposure

For each trial, the angular deviation between the target position and the movement endpoint is recorded. The rate of error reduction across trials represents the speed of adaptation.

After-effect

Following the exposure phase, open-loop pointing is used to assess the shift in perceived straight-ahead. The angular deviation from the true midline is taken as a measure of after-effect magnitude.

Transfer to spatial tasks

To evaluate whether the adaptation influences spatial representation beyond the trained movement, perceptual spatial tasks are administered immediately after exposure. Changes in spatial bias between pre- and post-exposure measurements indicate transfer of the recalibration.

3.1.4 Participants

The experiment is conducted with healthy adult participants. This allows controlled comparison of induction methods without the variability associated with clinical populations and is consistent with previous VR-based prism-adaptation studies[1, 6].

3.1.5 Apparatus and implementation

The study is implemented in an immersive virtual reality system with tracked hand or controller input. The sandbox architecture enables real-time manipulation of the spatial relation between physical movement and visual feedback and ensures that all induction methods are tested within the same virtual environment and task structure.

4.1 Task Decomposition of the Prism Adaptation Workflow

The experimental workflow consists of a *measurement–exposure–measurement* structure. A participant first performs a baseline measurement task, is then exposed to a visuomotor perturbation, and finally repeats the same measurement task in a post-exposure phase. In the current implementation, the primary measurement tasks are *Open-Loop Pointing* and *Line Bisection*. A *Landmark* task is retained as a future assessment task, primarily for potential clinical use. During exposure, one of four visual conditions is applied: *none*, *translation*, *rotation*, or *skew*.

At the output level, the tasks can be understood as follows:

- **Open-Loop Pointing:** discrete endpoint selection toward a single target, performed without online correction after commitment.
- **Line Bisection:** discrete spatial judgment of the perceived midpoint of a horizontal line, implemented as a pointing response.
- **Landmark:** comparative perceptual judgment, where the participant chooses which of two line segments appears longer.
- **Exposure:** repetitive target acquisition under a visual perturbation, used to induce sensorimotor adaptation.

4.1.1 Open-Loop Pointing

Purpose. Open-loop pointing is used as a discrete spatial measurement task. It estimates the participant's perceived target location before and after exposure by recording the signed pointing error relative to the intended target center.

Start posture and motor requirements. The participant begins each trial with the arms relaxed alongside the body. From this posture, the response arm is raised with a semi-straight configuration: the shoulder is flexed sufficiently to bring the hand into the frontal workspace, while the elbow remains only slightly flexed rather than fully bent. The intention is to maintain a stable, repeatable pointing posture rather than a crouched or highly retracted arm configuration.

Action sequence.

1. Lower the response arm to the start posture.
2. Raise the arm from the side into the pointing workspace.
3. Aim toward the single target location on the board.
4. Commit the response.
5. Lower the arm again before the next trial.

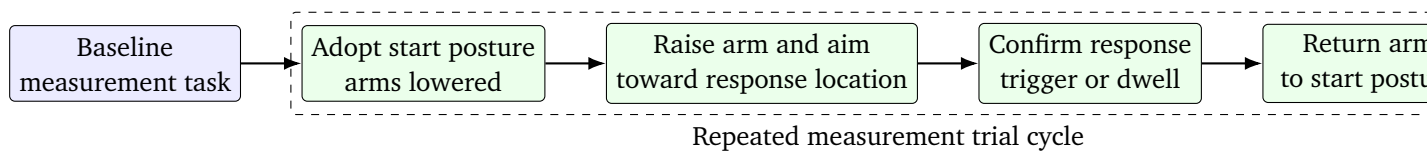


Figure 4.1: High-level workflow of a complete prism-adaptation run. The participant performs a baseline measurement task, an exposure block under a selected visual perturbation, and the same measurement task again in a post-exposure block.

Confirmation rule. The confirmation mechanism depends on input mode:

- **Controller mode:** the response is confirmed by trigger input.
- **Embodied mode:** the response is confirmed by maintaining the aim for approximately 1.5 s.

Feedback and feedforward.

- **Feedforward:** the visible target location, the board plane, and the participant’s prior expectation of where the target is located.
- **Feedback:** hit confirmation, the visible hit marker/readout, and the change from one accepted trial to the next.

Task characterisation. This is a discrete aiming task with a clear start posture, a ballistic approach toward a target, and a discrete response commitment. The key output is the endpoint of the selected response, not the continuous movement path itself.

4.1.2 Line Bisection

Purpose. Line bisection measures the participant’s perceived midpoint of a horizontal line. It is therefore also a discrete endpoint selection task, but the spatial judgment is no longer *where is the target?* but rather *where is the center of this line?*

Start posture and motor requirements. The start posture is the same as in open-loop pointing: the arm begins lowered at the participant’s side and is raised into a semi-straight pointing posture for each response.

Action sequence.

1. Lower the response arm to the start posture.
2. Raise the arm into the frontal workspace.
3. Inspect the presented horizontal line and estimate its midpoint.
4. Aim at the perceived midpoint.
5. Confirm the response.
6. Lower the arm before the next trial.

Confirmation rule. The confirmation rule mirrors open-loop pointing:

- **Controller mode:** trigger-based confirmation.
- **Embodied mode:** confirmation by maintaining the aim for approximately 1.5 s.

Feedback and feedforward.

- **Feedforward:** the visible line and the participant's internal estimate of the midpoint.
- **Feedback:** hit confirmation/readout and the resulting signed error relative to the true midpoint.

Task characterisation. This task combines a perceptual center estimate with a discrete pointing response. Compared with open-loop pointing, the motor structure is almost identical, but the perceptual demand is higher because the response location must be inferred rather than directly indicated.

4.1.3 Landmark Task

Purpose. The landmark task is primarily a perceptual assessment task. Rather than selecting a single spatial point, the participant decides which of two line segments appears longer. In the present project it is retained mainly for future clinical use and is less central than open-loop pointing and line bisection for healthy-participant experiments.

Action sequence.

1. Inspect the displayed landmark stimulus.
2. Compare the two visible line segments.
3. Decide which segment appears longer.
4. Select the corresponding side using the active input modality.

Feedback and feedforward.

- **Feedforward:** the visual structure of the landmark stimulus.
- **Feedback:** response confirmation and task progression to the next trial.

Task characterisation. Unlike the other two measurement tasks, landmark is not primarily a target-acquisition task. It is better described as a comparative perceptual judgment followed by a discrete categorical response.

4.1.4 Exposure Task

Purpose. The exposure task is the adaptation-inducing component of the workflow. The participant repeatedly acquires targets while a visual perturbation is applied. The available perturbations are *none*, *translation*, *rotation*, and *skew*.

Spatial layout. Three targets are arranged horizontally on the board plane: left, center, and right. One of these targets is selected pseudo-randomly on each attempt.

Motor structure. Unlike the measurement tasks, the participant does *not* need to lower the arms between attempts. Instead, the arm remains engaged in the task space and the participant continuously redirects the aim toward whichever target is currently active.

Action sequence.

1. Observe which of the three horizontal targets is currently active.
2. Move the pointing ray or fingertip toward that target.
3. Stabilise the aim within the target region.
4. Confirm the hit by the current confirmation rule.
5. Redirect toward the next randomly selected target without fully resetting the arm posture.

Confirmation rule.

- **Controller mode:** the default implementation uses trigger confirmation, although the system also supports Heisenberg-mitigation variants such as opposite-hand confirmation or controller hover confirmation.
- **Embodied mode:** the hit is confirmed through a dwell/hold gesture once the fingertip remains within the target region for the required duration.

Feedback and feedforward.

- **Feedforward:** active-target highlighting, the horizontal layout of the three targets, and the participant's expectation that a new target will appear after a successful hit.
- **Feedback:** hit/miss outcome, hit marker, attempt progression, and the currently applied perturbation as experienced through the altered visuomotor relationship.

Task characterisation. Exposure is a repetitive target-acquisition task with continuous visuomotor correction. In contrast to the measurement tasks, it is not primarily used to estimate a single perceptual response; it is used to drive adaptation through repeated target-directed movements under an imposed perturbation.

4.1.5 Summary of Task Properties**4.1.6 Interpretive Distinction Between Measurement and Exposure**

The most important conceptual distinction in the workflow is between *measurement tasks* and the *exposure task*. Open-loop pointing and line bisection are used to measure the participant's spatial response before and after perturbation. Exposure, in contrast, is the repetitive adaptation phase intended to alter that response. Landmark is currently orthogonal to this main baseline–exposure–post structure and is best interpreted as a supplementary assessment task rather than a primary adaptation measure.

Task	Primary output	Motor structure	Confirmation	Notes
Open-Loop Pointing	Signed endpoint error relative to a single target	Raise arm from side, point, confirm, lower arm again	Trigger in controller mode; dwell in embodied mode	Baseline/post measurement task emphasizing discrete endpoint selection
Line Bisection	Signed error relative to the true midpoint of a line	Same as open-loop, but response location is inferred from a displayed line	Trigger in controller mode; dwell in embodied mode	Baseline/post measurement task combining perceptual midpoint estimation with pointing
Landmark	Categorical left/right perceptual judgment	Observe stimulus, decide, select response side	Discrete response selection	Retained mainly for future clinical use
Exposure	Repeated hit/miss performance under perturbation	Continuous redirection among three targets without full reset between attempts	Trigger, hover-confirm, or embodied dwell depending on configuration	Adaptation-inducing task; perturbation active only in this phase

Table 4.1: Task decomposition summary for the implemented prism-adaptation workflow.

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Evaluation
